

PRESENT AND READY

Liveness Detection for Real-Time Face Recognition Attendance

USER MANUAL

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Department of Information Technology

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1. INTRODUCTION

Present and Ready is a web-based, AI-powered attendance system developed for classroom use at the College of Computing Studies (CCS), Western Mindanao State University (WMSU). It uses real-time face recognition combined with liveness detection to automatically record student attendance during class sessions — eliminating the need for manual roll calls, paper sign-in sheets, or RFID-based systems.

1.1 Purpose of This Manual

This manual provides step-by-step guidance for all users of the Present and Ready system. It covers registration, login, starting a class session, managing attendance records, generating reports, and troubleshooting common issues.

1.2 Intended Audience

User Role	Description
Administrator	Manages users, programs, academic structures, and system settings.
Sub-Administrator	Assists admin in managing schedules, instructors, and class sessions.
Instructor / Faculty	Starts class sessions, monitors live attendance, reviews and finalizes records.
Student	Registers biometric data and views personal attendance history.

Table 1: Intended Audience

1.3 System Overview

Present and Ready integrates multiple AI technologies to deliver a seamless attendance experience:

- YOLOv8 — detects faces and human bodies in real-time video frames
- InsightFace — generates a unique 512-dimensional facial embedding per student
- TorchReID (OSNet_x1_0) — continuously re-identifies individuals across video frames
- ONNX Liveness Model — distinguishes live faces from printed photos, videos, or screens
- Flask Web Application — serves the user interface and connects all modules
- MySQL Database — stores attendance records, embeddings, and user data

 **NOTE:** The system is designed for a single classroom of 10–15 concurrent students within a 60–70 sqm space, at an optimal camera distance of 1–3 meters.

2. SYSTEM REQUIREMENTS

2.1 Hardware Requirements

Component	Specification
Camera	Hikvision DS-2CD2086G2-I(U) 4K AcuSense Fixed Mini Bullet Network Camera (or equivalent ≥4K IP camera with RTSP support)
Processing Computer	Laptop/PC with at least Intel Core i5 (10th gen+) or equivalent; minimum 8GB RAM; dedicated GPU recommended
Network Connection	Stable Wi-Fi or LAN — 5 to 15 Mbps minimum for RTSP streaming and database access
Power Supply	UPS recommended for uninterrupted operation during power fluctuations

Table 2: Hardware Requirements

2.2 Software Requirements

Software	Details
Operating System	Windows 10/11 or Ubuntu 20.04+
Web Browser	Google Chrome (recommended), Mozilla Firefox, or Microsoft Edge — latest version
Python	Version 3.12 (for running the server locally)
MySQL	Version 9.3 or higher (cloud-based or local)
Cloudflare Tunnel	Required for remote/cloud access outside the local network

Table 3: Software Requirements

⚠ WARNING: The system requires an active internet connection at all times. Offline mode is not supported in the current version.

3. GETTING STARTED

3.1 Accessing the System

Open your web browser and navigate to the system URL provided by your institution's IT administrator. The login page will be displayed. If you are accessing the system for the first time, you must complete the registration process before logging in.

🔗 NOTE: The system URL is configured by the administrator. Contact your IT department if you do not have the link.

3.2 Registration Process

All users (instructors and students) must register before they can use the system. Follow the steps below:

Step 1 — Accept Terms and Conditions

Read the system's Terms and Conditions carefully. Check the agreement box and click Accept to proceed. This confirms your consent to biometric data collection as required by the Data Privacy Act of 2012 (RA 10173).

Step 2 — WMSU Gmail Verification

Enter your official institutional email address (format: studentid@wmsu.edu.ph for students, or your faculty email). Click Send OTP. Check your email inbox for a 6-digit One-Time Password.

Step 3 — OTP Verification

Enter the 6-digit OTP received in your email. Click Verify. The OTP is valid for a limited time — if it expires, click Resend OTP.

Step 4 — Fill Up Personal Information

Complete the registration form with your personal and academic details:

- Full name (Last, First, Middle Initial)
- Student ID or Employee ID
- Program and year level (for students)
- Contact number and department

Click Next to proceed.

Step 5 — Biometric Capture (Face Encoding)

This is the most critical step. The system will prompt you to:

1. Upload a clear, front-facing selfie photo
2. Complete a live face scan using your device's camera

 **NOTE:** For best results: ensure good lighting (no backlighting), face the camera directly, remove glasses or face coverings, and keep a neutral expression. Poor-quality images will result in recognition errors.

Step 6 — Registration Summary

Review all submitted information on the summary page. If everything is correct, click Submit Registration. Your account will be pending admin approval.

⚠ WARNING: Your account is NOT active until the Administrator reviews and approves your registration. You will receive an email notification once approved.

3.3 Logging In

Once your account is approved:

3. Go to the system URL and click Login
4. Enter your registered email address and password
5. Click Sign In

You will be redirected to your role-specific dashboard.

3.4 Password Recovery

If you forget your password:

6. On the Login page, click Forgot Password
7. Enter your registered email address and click Send Reset Link
8. Check your email and enter the OTP code received
9. Enter and confirm your new password, then click Save

4. ADMINISTRATOR GUIDE

The Administrator has full control over the system, including user management, academic structure configuration, and system oversight.

4.1 Admin Dashboard

Upon login, the Admin Dashboard displays a real-time overview of the system including total registered students, faculty members, and program-specific counts (CS, IT, ACT). Quick navigation links provide access to all management modules.

4.2 Managing Academic Structure

Before any class sessions can be run, the admin must set up the following in order:

#	Module	Description
1	Academic Year	Create and manage official academic years (e.g., 2025–2026).
2	Semester	Set up First Semester, Second Semester, and Summer terms.

#	Module	Description
3	Program	Add academic programs (BS Information Technology, BS Computer Science, etc.).
4	Curriculum	Define curriculum structure per program and year level.
5	Subject	Create and categorize subjects/courses.
6	Class Schedule	Assign subjects, instructors, days, and time slots to classes.

Table 4: Managing Academic Structure

4.3 User Management

Approving Registrations

- Go to Pending Registrations in the admin panel
- Review the submitted information and biometric data
- Click Approve to activate the account, or Reject with a reason

Managing Faculty and Students

- Navigate to Faculty List or Student List
- You can view, edit, deactivate, or delete user accounts
- Use the search and filter tools to find specific users

4.4 Attendance Rules Configuration

Attendance status categories and their thresholds are configured by the admin. The system supports five statuses:

Status	Description
Present	Student recognized within the first few minutes of class, before the late threshold.
Late	Student recognized after the configured late threshold but before the session ends.
Missing	Student was detected at some point but could not be re-verified.
Excuse	Manually applied by instructor for students with valid excused absences.
Absent	Student was not detected during the entire class session.

Table 5: Attendance Rules Configuration

4.5 Sub-Administrator Assignment

Admins can designate trusted users as Sub-Administrators. Sub-admins can assign instructors to subjects, manage schedules, and initiate class sessions, but cannot access full system settings or delete user accounts.

5. INSTRUCTOR / FACULTY GUIDE

Instructors use the system to start and manage class sessions, monitor live attendance, review records, and generate reports.

5.1 Instructor Dashboard

The Instructor Dashboard shows assigned classes and schedules, recent attendance activity, and quick access to start a session or view reports.

5.2 Starting a Class Session

Follow these steps to begin recording attendance for a class:

Step 1 — Go-To Timer Setup

From your dashboard, click Start Class Session or navigate to the Go-To Timer page. Configure the following:

- Select the RTSP camera feed for your classroom
- Choose the class from your assigned schedule
- Set the session duration (e.g., 1 hour, 1.5 hours, 3 hours)
- Set the late threshold (e.g., 15 minutes — students who arrive after this are marked Late)

 **NOTE:** Make sure the camera is powered on and connected to the network before starting. Verify the RTSP stream is active.

Step 2 — Launch the Live Session

Click Start Session. The system will:

10. Connect to the RTSP camera stream
11. Begin face detection and recognition in real time
12. Display a live feed with bounding boxes around detected faces
13. Show recognized student names and their attendance status on the dashboard

Step 3 — Monitor Live Attendance

During the session, the Live Class Session UI shows:

- The real-time camera feed with detection overlays
- A running list of recognized students and their status
- Flagged or unknown faces for your review
- Session timer countdown

⚠ WARNING: If a student is not being recognized, check that they are within 1–3 meters of the camera, facing forward, and in good lighting. They may need to re-register their biometric data if recognition consistently fails.

Step 4 — End the Session

Click End Session when class is over. The system will:

14. Finalize all attendance records
15. Generate a session summary showing the complete attendance list
16. Automatically save records to the MySQL database and a CSV backup

5.3 Reviewing and Editing Attendance

After a session, you can review the attendance records:

17. Navigate to Attendance Records and select the class and date
18. The system displays each student's status with a timestamp
19. To edit an entry, click the Edit icon next to the student's name
20. Change the status (e.g., from Absent to Excuse) and add a reason note
21. Click Save to confirm the change

🔗 NOTE: All manual edits are logged for audit purposes. Edits cannot be made after the instructor finalizes the record.

5.4 Finalizing Attendance Records

Once you have reviewed and corrected all entries, click Finalize Attendance. Finalized records are locked and cannot be edited. These records are then available for administrative reporting.

5.5 Generating Reports

To export attendance data:

22. Go to Reports from your dashboard
23. Select the class, date range, and report format (CSV or Excel)
24. Click Generate Report
25. Download the file to your device

6. STUDENT GUIDE

As a student, you do not need to take any active steps during a class session — the system recognizes you automatically. Your responsibilities are to register properly and maintain your biometric data.

6.1 Student Dashboard

Your dashboard displays your personal attendance summary for all enrolled classes. You can view your attendance history, current status per subject, and any system notifications.

6.2 Attending Class

When your instructor starts a class session, simply:

26. Arrive at the classroom and sit within 1–3 meters of the camera
27. Face the camera naturally (you do not need to stand up or pose)
28. The system will automatically detect and recognize your face
29. Your attendance will be logged — no action needed from you

 **NOTE:** Avoid covering your face with masks, hats, or large accessories during the scanning period at the start of class for best recognition results.

6.3 Updating Biometric Data

If you experience consistent recognition failures, you can update your face data:

30. Log in to your dashboard
31. Navigate to Profile > Update Biometric Data
32. Upload a new clear selfie or redo the live face scan
33. Submit for admin approval

 **WARNING:** Biometric updates require admin approval before taking effect. Submit well before an important class session.

6.4 Viewing Attendance Records

To check your attendance:

34. Log in to your dashboard
35. Click on a subject to see your attendance history
36. Status labels include: Present, Late, Missing, Excuse, Absent
37. If you believe an attendance entry is incorrect, contact your instructor directly

7. LIVENESS DETECTION EXPLAINED

Liveness detection is the system's anti-spoofing mechanism. It ensures that the face being recognized belongs to a real, live person — not a printed photo, a video replay, or a digital screen.

7.1 How It Works

The system uses a combination of methods to verify liveness:

Method	Description
ONNX Model (model_rgb.onnx)	Analyzes texture and depth patterns to detect spoofed faces.
Eye Aspect Ratio (EAR)	Detects natural blinking patterns using facial landmark analysis.
Mouth Aspect Ratio (MAR)	Detects subtle mouth movements that are absent in static spoofs.
Face–Body Association	Matches a detected face with a corresponding human body to prevent floating face attacks.

Table 6: How It Works

7.2 Spoof Prevention Rules

The system enforces 10 heuristic rules to prevent false attendance. Key rules include:

- A face must be continuously visible for a minimum number of frames before recognition is attempted
- Recognition only proceeds after liveness is confirmed
- Each student can only be logged once per session (no duplicate entries)
- Faces without a detected body are flagged and excluded
- Unknown faces are isolated and do not affect attendance records

8. TROUBLESHOOTING

Problem	Possible Cause	Solution
Student not recognized	Poor biometric registration or out-of-range position	Re-register biometric data; ensure student is 1–3m from camera
Bounding box jumping to wrong person	Another student is blocking the recognized user	Ensure students are seated with clear lines of sight to camera
Liveness check failing for real person	Low lighting or face partially covered	Improve lighting; remove face coverings; re-register if needed
Session not starting	Camera not connected or RTSP URL incorrect	Verify camera power and network; confirm RTSP URL in settings

Problem	Possible Cause	Solution
Slow frame rate / lag	Weak internet connection or high CPU load	Check network speed; close unnecessary applications on the server
OTP not received	Email delay or spam filter	Check spam/junk folder; wait 2 minutes then request a new OTP
Attendance not saving to database	Network interruption during session	Attendance is backed up in CSV; sync when connection is restored
Wrong student marked as present	Weak face encoding from registration	Have the student re-register with better-quality photos

Table 7: Troubleshooting

9. PRIVACY AND DATA SECURITY

Present and Ready is designed in full compliance with the Data Privacy Act of 2012 (Republic Act No. 10173) of the Philippines. The following measures protect all user data:

Measure	Details
Biometric Storage	Facial data is stored as numerical embeddings only — raw images are not permanently retained.
Password Security	All passwords are hashed using bcrypt and never stored in plain text.
Access Control	Role-based access ensures users only see data relevant to their role.
Session Security	Secure UUID session tokens prevent unauthorized access.
No Third-Party Sharing	Biometric and attendance data is never shared with third parties.
Data Disposal	Embeddings and records are removed upon request or study termination.
Encrypted Database	MySQL database access is restricted to authorized administrators only.

Table 8: Privacy and Data Security

Students and faculty have the right to:

- Access their personal data stored in the system
- Request correction of inaccurate records
- Request deletion of their biometric data upon leaving the institution

 **NOTE:** For data privacy concerns or requests, contact the system administrator or the WMSU Data Protection Officer.

10. QUICK REFERENCE CARD

Administrator Quick Steps

- 38. Setup: Academic Year → Semester → Program → Curriculum → Subject → Schedule
- 39. Approve/manage user registrations from the Pending Registrations page
- 40. Assign Sub-Admin roles as needed

Instructor Quick Steps

- 41. Log in → Dashboard → Start Class Session
- 42. Configure timer and late threshold → Launch live session
- 43. Monitor attendance in real time → End session
- 44. Review and edit records → Finalize → Export report

Student Quick Steps

- 45. Register → Verify email → Upload selfie + complete face scan → Wait for approval
- 46. Sit within 1–3m of camera during class → System records attendance automatically
- 47. Check dashboard for attendance history

Key Performance Benchmarks

Metric	Target / Achieved
Face Detection Accuracy	≥85% (Achieved: 85.2%)
Face Recognition Precision	≥85% (Achieved: 85%)
Liveness Detection Accuracy	≥90% (Achieved: 89.8%)
Average Response Latency	≤10.35 seconds (Achieved: 10.35 sec)
User Satisfaction (ISO/IEC 25010)	≥80% (Achieved: 4.56/5 = 91.2%)
Optimal Camera Distance	1–3 meters
Supported Classroom Size	Up to 15 concurrent students, 60–70 sqm

Table 9: Key Performance Benchmarks

11. SUPPORT AND CONTACT

For technical support, system issues, or data privacy requests, please contact the following:

System Developers	Berioso, Rhodmin Lou B. Quimba, Guilmar A. Tilde, Lawrence Olivier N.
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Table 10: Support and Contact

For the Head of Department or Dean's Office, please refer to your institutional directory.

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